7. Construction Materials Catalog & Engineering Standards

Document ID: DOC-07-MAT | **Version:** 6.0 (2026) | **Department:** Structural Engineering & QA/QC
Primary OWL Classes: `Material`, `Concrete`, `Brick`, `Steel`, `Wood`, `Insulation`, `Glass`
Primary OWL Properties: `usesMaterial`, `suppliesMaterial`, `owl:disjointWith`

7.1 Taxonomy of Construction Materials in the Ontology

The top-level class `Material` represents all physical construction materials consumed by a `Project` (via object property `usesMaterial`) or handled by specialized workers (`Electrician` and `Plumber` both satisfy `usesMaterial some Material`) and delivered by a `Supplier` (via object property `suppliesMaterial`). In `stavebna_firma_ontology - kópia.owl`, `Material` is disjoint with `ConstructionCompany`, `Person`, `Project`, `ConstructionMachine`, and `ConstructionPhase`.

The ontology defines exactly **six direct subclasses of `Material`**:

1. **`Concrete` (Betón):** Structural ready-mix and precast concrete (EN 206).
2. **`Brick` (Tehla):** Fired clay masonry blocks, acoustic bricks, and ceramic lintels (EN 771-1).
3. **`Steel` (Oceľ):** Structural carbon steel profiles (S235, S355) and reinforcing rebar (B500B).
4. **`Wood` (Drevo):** Structural solid timber (C24), glued laminated timber (Glulam GL24h/GL28h), and Cross-Laminated Timber (CLT).
5. **`Insulation` (Izolácia):** Thermal, acoustic, and waterproofing materials (stone mineral wool, EPS/XPS, PIR boards).
6. **`Glass` (Sklo):** Architectural float glass, tempered/laminated safety glazing, and triple-glazed curtain wall units.

7.2 Formal OWL Disjointness & Restriction Rules on Materials

> **Critical Ontology Rules for `Concrete`:**
>
> - **Disjointness Axiom 1:** `Concrete owl:disjointWith Wood` (No material instance can simultaneously be both `Concrete` and `Wood`).
> - **Disjointness Axiom 2:** `Concrete owl:disjointWith Glass` (No material instance can simultaneously be both `Concrete` and `Glass`).
> - **Supplier Restriction on Concrete:** `Concrete ⊑ suppliesMaterial some Supplier` (Every batch of `Concrete` in the ontology is linked to a certified `Supplier`).

Why Material Disjointness Matters in Reasoning: Consider the class `WoodenHouse`, which is restricted by `usesMaterial only Wood` (`owl:allValuesFrom Wood`). Because `Concrete` is disjoint with `Wood`, any project that uses `Concrete` (such as `EuroveaTower` or `FamilyHouseExample`) is mathematically proven by the OWL reasoner not to be a pure `WoodenHouse`.

7.3 Technical Specifications & Project Usage Matrix

Material Subclass	Standard Grades Used by ApexBuild	Key Physical Properties	Explicitly Linked Projects / Classes in Ontology

`Concrete`	C25/30 XC2 (foundations), C35/45 (columns), C60/75 (high-rise cores)	Compressive strength 30–75 MPa; density 2,400 kg/m³	`EuroveaTower` (`usesMaterial some Concrete`), `FamilyHouseExample` (`usesMaterial some Concrete`)
`Brick`	Porotherm 38 Ti Profi, Porotherm 30 AKU (acoustic)	Thermal conductivity λ = 0.079 W/(m·K); fire class A1	`FamilyHouseExample` (`usesMaterial some Brick`), historical masonry reconstructions
`Steel`	Structural profiles S355J2 (EN 10025); Rebar B500B (EN 10080)	Yield strength 355–500 MPa; modulus of elasticity 210 GPa	`EmpireStateBuilding` (`usesMaterial some Steel`), commercial office frames, industrial trusses
`Wood`	Spruce C24, Glulam GL24h, 5-layer CLT panels	Bending strength 24 MPa; density 420–480 kg/m³; renewable carbon sink	`WoodenHouse` (`usesMaterial only Wood`)
`Insulation`	Rockwool Frontrock MAX E (mineral wool), Austrotherm XPS TOP 30	λ = 0.035 W/(m·K); fire resistance class A1 (mineral wool)	Used across all `ResidentialProject`, `CommercialProject`, and `FinishingWork` phases
`Glass`	Guardian SunGuard Triple IGU (44.2 / 16 Ar / 6 / 16 Ar / 44.2)	U_g = 0.5 W/(m²K); acoustic attenuation R_w = 43 dB	High-rise facades on `EuroveaTower` and modern `OfficeBuilding` curtain walls

7.4 On-Site Quality Control & Material Testing Protocols

Every delivery of `Material` to an ApexBuild `Project` undergoes mandatory inspection before incorporation into `Foundation`, `StructuralWork`, or `FinishingWork`:

- **`Concrete` Slump & Cube Testing:** For every 50 m³ poured, site technicians cast three 150x150x150 mm test cubes for 7-day and 28-day destructive compression testing in an accredited laboratory.
- **`Steel` Mill Certificates & Weld NDT:** All structural `Steel` requires an EN 10204 Type 3.1 inspection certificate. Load-bearing welds undergo 100% ultrasonic testing (UT).
- **`Wood` Moisture Verification:** CLT and Glulam timber elements delivered for a `WoodenHouse` must have a measured equilibrium moisture content of 12% ± 2% prior to assembly.